

How frequently is psychological reactance on TikTok, and how quickly can it be detected?

A benchmark on 1200 comments under videos of German politicians, with two codings, two conditions and four models

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Abstract

Public concern about political TikTok comment sections suggests that psychological reactance is common. Two benchmarks show the opposite. On 1200 party-stratified comments, four models coded between about 2.8 and 10.8% as reactance under strict, theory-anchored coding; the newest backend, GLM-5.3-Flash, is the least conservative (9–11%). A second, comment-only pass over 2001 comments yields 2.90% (95% CI [2.20; 3.60], Jev). The video transcript does not measurably improve aggregate detection – and the case-level confusion analysis shows it does not change which individual comments are called reactant either. Jev, a Decision-API backend, answers in 0.46 s and costs \$0.061 per 1,000 comments and *as the only backend* returns class probabilities, so a threshold lifts consensus-agreement precision from 14% to nearly 80% at $t=0.6$. The real bottleneck is validity: a manual re-coding of the 36 audited three-model positives found 46% precision overall (92% at three-model consensus, 25% for single-model flags; exact Wilson intervals on $n=12$ per stratum). GLM-5.3-Flash flags 61 comments no other model calls reactant; the audit predates GLM, so their precision is an open question, not a finding. Two sharpenings of the type codebook derived from that audit – the trigger must be the target of the reaction, and politeness markers neither create nor cancel reactance – raise the A/B codebook agreement on the large-scale sample to raw 98.2% ($\kappa = 0.98$). A gating variant (Section 5) runs a cheap Jev binary pass first and classifies the type, across all four models, only on the $\approx 3\%$ it lets through; separating the gate question (do all models agree reactance exists?) from the conditional six-class type question shows that a substantial share of the apparent disagreement concerns the gate premise, not the type.

Keywords: psychological reactance; social media; LLM coding; TikTok; validity; reliability; bootstrap; decision API

1 Data and method

Samples. Two samples from the same corpus of German-language comments under the accounts of German politicians, stratified by the account’s party. The *matrix sample* comprises 1200 comments from 68 accounts, 16 parties and 400 videos, each with its video transcript (required by Condition A). The *large-scale sample* adds 2001 comments from 114 accounts and 16 parties, coded exclusively under Condition B (comment text only) and, for the agreement analyses, by the two backends that run on that sample (Jev and GLM-5.3-Flash). Both samples are restricted to visible top-level comments of at least 25 characters.

Codebook A (binary). Requires, together, a perceived freedom threat *and* an affective or behavioural reaction. The project’s decoding perspective governs: what matters is not whether a message was actually controlling but whether the commenter perceived it that way.

Codebook B (type). Seven disjoint labels, condensed from the project’s eight written-reactance archetypes (Appendix B). The first version lacked the link to the freedom threat; the models then coded ordinary political criticism as reactance (29–56%). Two gate fixes addressed this (Sections 3, 4).

Conditions and models. Condition A presents the comment *with* the video transcript; Condition B the comment only. Jev runs via the Decisions API, GPT-6-Luna, DeepSeek-V4.1-Flash and — new in this version — GLM-5.3-Flash via chat completion. All four receive identical instructions at temperature 0 with a fixed seed; GLM-5.3-Flash was the addition of this iteration and is otherwise handled exactly like the other chat models.

Volumes. The matrix is 2 codebooks $\times$ 2 conditions $\times$ 4 models = 16 calls per comment (1200 comments). This iteration added the fourth model: 4,800 GLM calls on the matrix sample plus 340 further calls to complete GLM on the large-scale sample ($\approx$ \$0.69). The two validation experiments (1,360 calls, \$0.075) and the Jev-only gate check are unchanged. The gate condition (Section 5) reuses the predictions already on disk and therefore costs no API calls. Total cost 3.17 US\$.

2 Frequency and speed

Reactance is rare. Table 1 reports prevalence with 95%-bootstrap confidence intervals on the matrix sample. The rank order is stable across conditions; Jev marks least often.

Model	Cond	n	Prev.	95% CI
jev-1.13	A	1200	3.83%	[2.83; 5.00]
gpt-6.luna	A	1200	6.58%	[5.25; 8.00]
deepseek-fl.	A	1200	6.17%	[4.83; 7.50]
glm-5.3-fl.	A	1200	10.83%	[9.08; 12.67]

Table 1. Prevalence, Codebook A, with 95%-bootstrap CI (2,000 resamples), matrix sample.

Jev remains the fastest and cheapest backend: 0.46 s and \$0.061 per 1,000 comments, against 2.1–2.4 s and \$0.12–0.33 for the chat models (Figure 3); GLM-5.3-Flash is the cheapest chat model (\$0.119) but does not beat Jev. Jev is also the only backend that returns class probabilities — an advantage with a practical payoff in Section 4.4 (a threshold on the probability lifts consensus-agreement precision from 14% to nearly 80%).

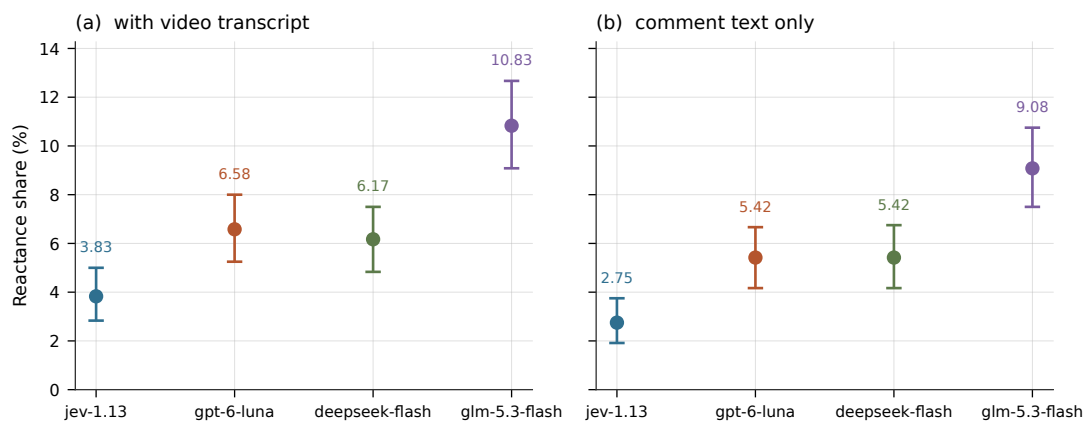


Figure 1. Prevalence by codebook and condition with 95%-bootstrap CIs.

2.1 Does the video context help detection?

Dropping the transcript moves the *aggregate* prevalence by less than two percentage points, in the same direction for all models (Δ prevalence, Codebook A, matrix sample: jev 1.1 pp, gpt-6-luna 1.2 pp, deepseek 0.8 pp, glm 1.8 pp; only Jev’s shift reaches the conventional significance level of an exact McNemar test). The aggregate figure alone does not answer the interesting question, however, which

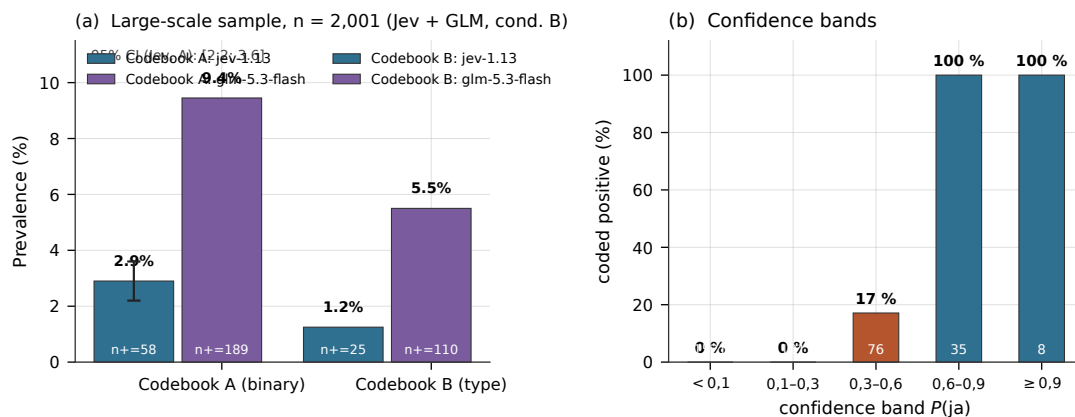


Figure 2. Large-scale run: prevalence of both codebooks (a) and the confidence-band structure of the Decisions API (b).

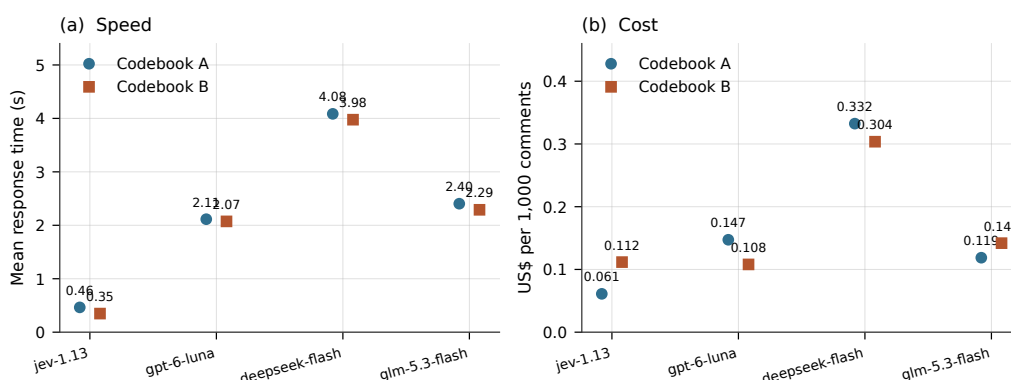


Figure 3. Response time (a) and cost (b) per 1,000 comments, Condition A. Circles: Codebook A. Squares: Codebook B.

is whether the *same comments* are classified as reactant with and without the transcript (Figures 4 and 5 sit here on purpose). Case-level, the picture is the same in substance: for Jev, 1173 of 1200 comments are classified identically in both conditions (case-level agreement 97.75%), and only 27 individual comments flip — 7 gain the reactance label, 20 lose it. The same pattern holds for the other models: flips range from 11 to 59 on Codebook B and from 27 to 121 on Codebook A (Figures 4–5) — double-digit flip counts in every case, but in absolute terms a small share of the sample. Small prevalence shift and small case-level flip shares are both observed; the transcript contributes neither aggregate nor item-level detection value. The transcript is therefore not a detection tool, but the comment text is sufficient for detection.

2.2 GLM-5.3-Flash is the least conservative model

The fourth model changes the prevalence picture more than any transcript or codebook change did. GLM-5.3-Flash marks 10.8% of the matrix sample as reactance (Condition A) — two to three times the share of the other three (2.8–6.6%) — and flags 189 of the 2001 large-scale comments where Jev finds 58. This is not better or worse recall in itself; it is the single largest source of the disagreement structure in Section 3, and it is a statement about *agreement with the consensus*, not about which model is correct: GLM deviates most strongly from the four-model consensus, which is exactly what its lower consensus-F1 measures (0.52–0.54, Table 2).

Of the comments only GLM flags, 61 were flagged by no other model in the matrix sample. The manual audit (Section 4.1) was drawn *before* GLM existed and says nothing about this GLM-only stratum; the v2 audit sample (48 candidates, the GLM-only stratum included) is drawn and awaiting

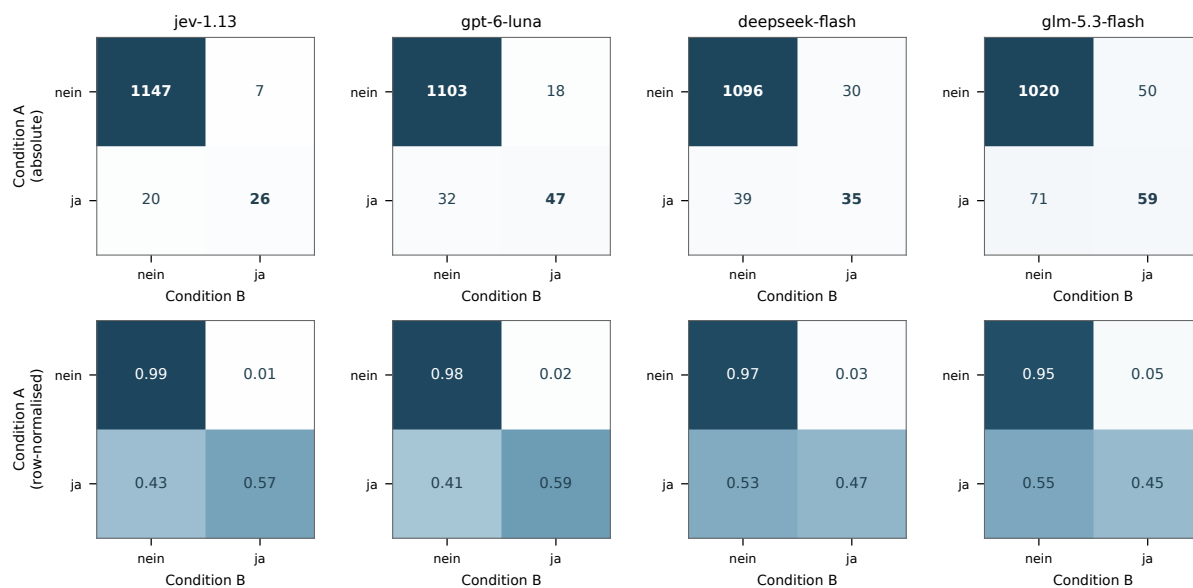


Figure 4. Condition A (with transcript) vs. Condition B (comment only), Codebook A: case-level stability of the binary reactance label per model. Rows = Condition A, columns = Condition B.

annotation. No precision claim about GLM-only flags is made here.

3 Agreement between models

With the fourth model in the matrix, the picture sharpens. The three original models still agree to 94.6–95.1% raw on Codebook A, Condition A, while GLM-5.3-Flash sits lower: 90.5–91.8% against each of them, because it marks roughly twice as many comments as reactance (10.8% prevalence, the highest of the four; Section 2). Yet even there the chance-corrected coefficients tell a two-part story: Cohen’s κ is low (0.31–0.59 across all pairs) and Gwet’s AC1 high (0.89–0.94). This is not a contradiction but a base-rate artefact: with 3–11% positives all models agree on the easy majority while the few positives split, and κ penalises exactly that. **For these data AC1 — and, for the binary task, the per-model F1 of Section 4.2 — are the appropriate measures, not κ alone.**

All of the following numbers — raw agreement, the two chance-corrected coefficients and F1 — lie on the *same* 0–1 scale. Raw agreement is the same proportion that the older version plotted as 94.6% on a separate percentage axis; expressing it as 0.946 is what lets one axis carry every quantity and makes the base-rate gap readable at a glance (Figure 6).

Model	<i>n</i>	Raw	F1	κ	AC1
deepseek-fl.	1200	0.977	0.774	0.762	0.974
glm-5.3-fl.	1200	0.928	0.522	0.492	0.917
gpt-6.luna	1200	0.971	0.729	0.714	0.968
jev-1.13	1200	0.973	0.667	0.653	0.971

Table 2. Each model against the four-model majority vote (Codebook A, Condition A). All four columns sit on the 0–1 scale — raw agreement and F1 next to the two chance-corrected coefficients, so F1 is a first-class measure here rather than a chapter of its own. The reference is the consensus, not a gold standard: these are agreement numbers, not correctness numbers.

4 Validity: the real bottleneck

A prevalence figure is only as solid as the coding rules behind it. Three checks address this.

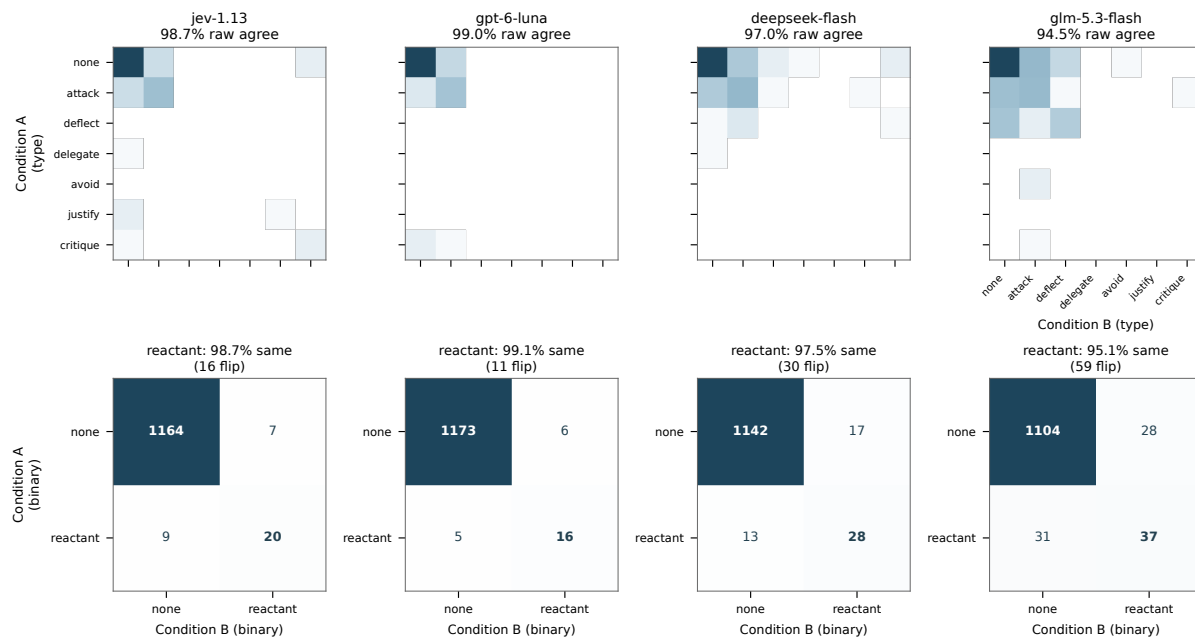


Figure 5. Condition A vs. Condition B, Codebook B. Top row: the full 7-class confusion — does the transcript change the inferred *kind* of reactivity? Bottom row: the binary reactant/none collapse with the four cells — are the *same comments* called reactant?

4.1 Precision of positive detection

Of the 119 comments marked as reactant by at least one of the *three original models*, 36 were re-coded manually, stratified by consensus degree (12 per stratum). **Scope of this audit: the three original models.** GLM-5.3-Flash was added to the experiment after the sample was drawn, so the strata below are “*k* of the three original models” and the audit says nothing about GLM-only flags. The estimated precision is 46% (stratum-weighted over the 3-model pool of 119): 92% (11/12 of the cases) at three-model consensus, 50% (6/12) at a two-model majority and 25% (3/12) for single-model flags; every value is a point estimate on $n = 12$ and the intervals are exact Wilson intervals (Figure 9). The error is structured: outrage without a freedom reference; reaction to a claim rather than to a constraint; named triggers without reactive behaviour.

Because most of the 3-model positives are single-model flags, a pipeline that runs a single model inherits roughly two thirds false alarms. The prevalence figures in Section 2 are therefore upper bounds; the order of magnitude survives, the exact percentages do not. A v2 audit sample covering the four-model pool (including the 61 GLM-only flags) is drawn and awaiting annotation; precision for the 4-model strata is a follow-up, not a number in this report.

4.2 F1 against the consensus majority

F1 is treated as a first-class agreement measure in Section 3, where it sits next to raw agreement, κ and AC1 (Table 2) rather than in a chapter of its own. The reference there is the four-model majority vote — not a gold standard — so F1 reads as how far a model sits from the consensual position, not how correct it is. At low prevalence precision and recall pull against each other: Jev has the best precision but gives up the most recall, and so loses to the chat models on F1.

4.3 Codebook A and Codebook B agree — on both samples

The error patterns above were translated into a new codebook, together with the surface-robustness finding (Section 6). Gate 2 requires the trigger to be the *target* of the reaction, not its topic: responding to a claim is not reacting to a constraint. Gate 3 fixes that politeness markers are neither trigger nor

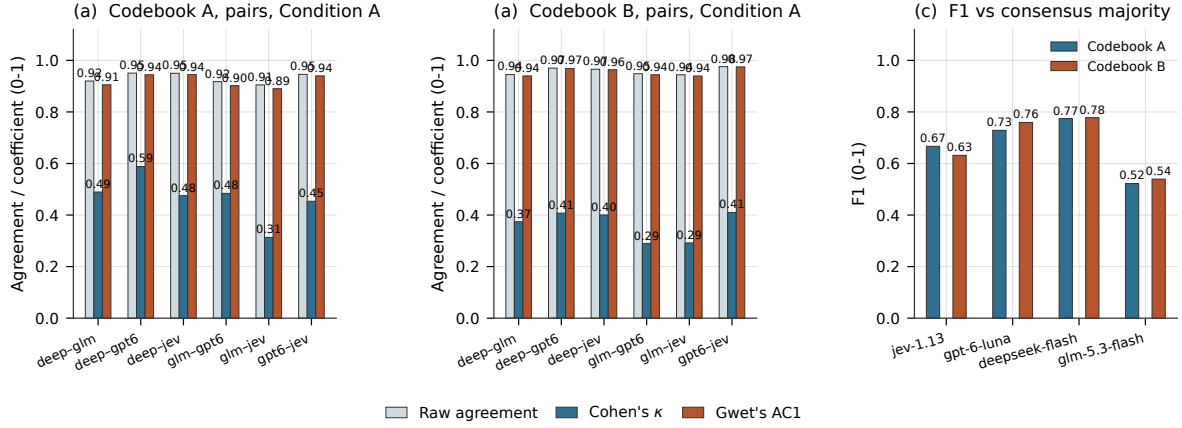


Figure 6. Pairwise agreement on the model pairs (a, b) and per-model F1 against the four-model majority (c), all on a single 0–1 axis: raw agreement, Cohen’s κ , Gwet’s AC1 and F1. The gap between raw agreement (≈ 0.91 – 0.95) and κ (0.29 – 0.59) is the base-rate artefact; AC1 and, for the binary task, F1 close it.

shield. Both gates sit in the chat instructions *and* in the Jev criteria, because the Decisions API only returns the criteria.

Two samples, two tables — deliberately kept apart, because mixing them is what the earlier draft did. On the *large-scale* sample (2001 comments, Condition B, Jev and GLM) the two instruments now agree to 98.2% raw with $\kappa = 0.98$ and an FP/TP ratio of 0.09 (Jev; Table 3). The *matrix sample* (1200 comments, all four models) shows the same pattern per model and condition at $n=1200$ (Table 4); its aggregate raw agreement is lower simply because the gate is tested on the harder, transcript-bearing comments.

Model	n	both	FP	FN	raw agree	κ	FP:TP
glm-5.3-fl.	2001	73	37	116	92.3	0.92	0.51
jev-1.13	2001	23	2	35	98.2	0.98	0.09

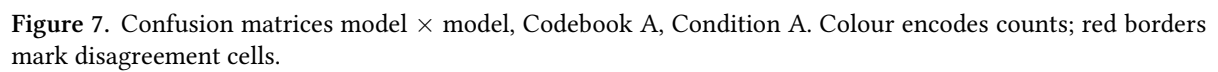
Table 3. Codebook A $\times$ B agreement, *large-scale sample* (2001 comments, Condition B). The n column is part of the point: these rows are 2,001, not the 1,200 of the matrix sample.

Model	n	both	FP	FN	FP:TP
deepseek-fl.	1200	35	10	30	0.29
glm-5.3-fl.	1200	42	23	67	0.55
gpt-6.luna	1200	19	3	46	0.16
jev-1.13	1200	17	10	16	0.59

Table 4. The same agreement, *matrix sample* (1,200 comments, Condition B, all four models).

4.4 Confidence as a corrective

Jev returns a class probability $P(\text{ja})$ for every comment; the other backends do not. Two properties of that number are worth checking separately, and Figure 11 shows both. We are careful about what this establishes: there is *no independent human-labelled calibration set* in this experiment, so neither panel is a calibration result in the statistical sense. The reference in both panels is the four-model consensus majority — a proxy, not ground truth — and the correct reading is **probability alignment with the model consensus**, with probability *discrimination* (do the probability bands separate consensus



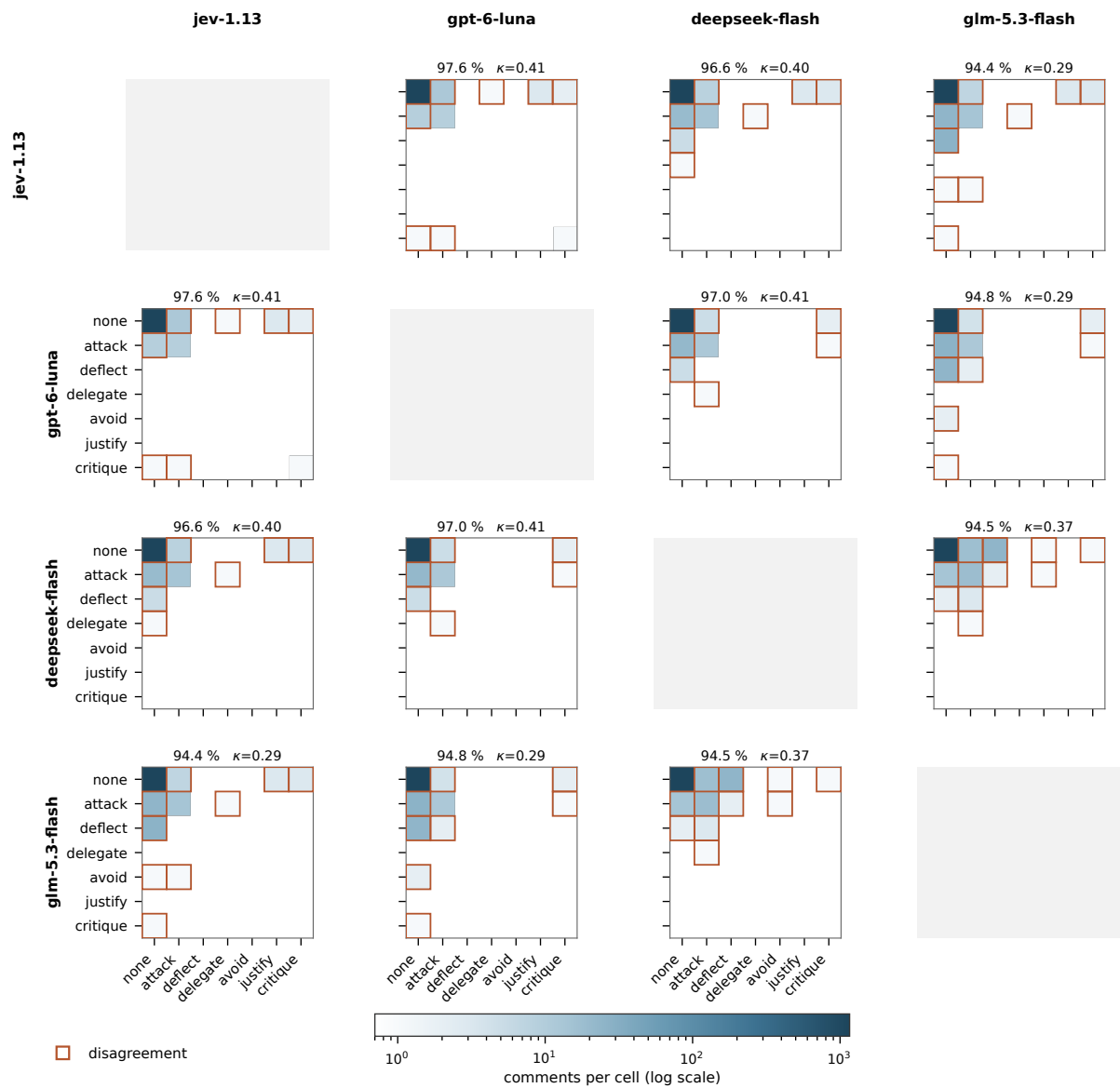


Figure 8. The same matrix for Codebook B. Cells carry colour only (no numbers): the 7-class structure is read visually; values appear in the appended tables.



Figure 9. Precision by consensus degree (a) and the distribution of the 3-model positives (b); the audit is explicitly the three original models (GLM was added later and is not covered). Error bars: exact Wilson 95% intervals on $n = 12$ per stratum.

positives from non-positives?) being the property that *is* demonstrated. Brier scores and ECE against human labels require a labelled validation set and are deliberately not reported here.

(a) Alignment against the consensus majority. We split Jev’s predictions into ten bins of $P(\text{ja})$ and, in each bin, ask how often the comment’s label agrees with the four-model majority vote of the other backends. If Jev’s probability were well *aligned* with the consensus, the observed rate would track the diagonal: predictions of $P = 0.6$ would be majority-reactant about 60% of the time. Jev is close across the range – the observed rate tracks the diagonal from $P < 0.1$ (observed 1%) up to $P > 0.9$ (observed 100%) – with a slight dip in the 0.3–0.6 bins, where a fifth to a third of its positives are either missed or refuted by the others. Alignment, then, is imperfect in exactly the region that matters at low prevalence.

(b) The threshold trade-off. Because $P(\text{ja})$ is meaningful, we can discard low-confidence flags. Raising the threshold t – flag only those comments with $P(\text{ja}) \geq t$ – trades coverage for consensus-agreement precision: at $t = 0.05$ nearly half the comments are flagged and precision is only 14%; at $t = 0.6$ precision reaches 79% but covers only 2.8% of the sample. The 46% line marks the precision of Jev’s raw threshold-free label. Precision, then, is not a property of the model but of the operating point chosen on this curve – relative to the consensus, not to a ground truth.

5 The gated condition: classify the type only when Jev says yes

Codebook B’s seven-way type classification is the expensive part of the pipeline. A gating variant answers the binary question first (Codebook A, Jev only) and runs the type classification only on the comments the gate lets through. Every model – Jev included, all four – then classifies the type of reactance on exactly those comments, so the four annotate the same set and the matrix of confusion matrices is directly comparable. It costs no additional API calls in this benchmark, because the Codebook B predictions were already collected: the gate is a recombination of existing labels.

This section separates two questions that the previous draft answered with one figure, and the separation is the substantive point.

Step 1 – detection. Jev performs binary reactance detection over all comments. On the matrix sample, condition B, the gate lets through 33 of 1200 comments (2.75%); on the large-scale sample 58 of 2001 (2.90%).

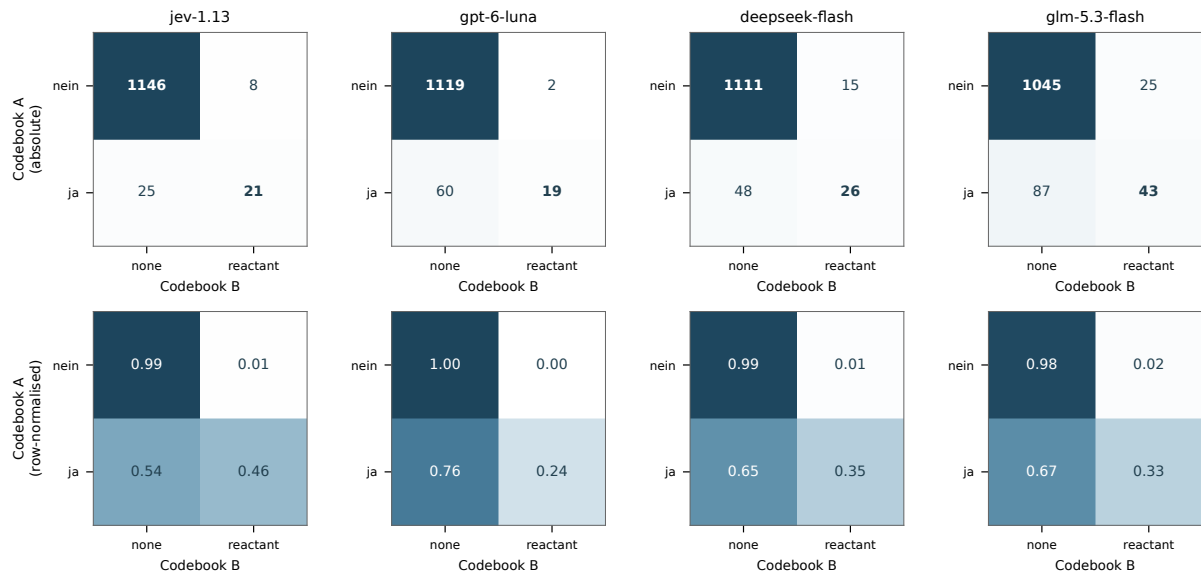


Figure 10. Confusion of Codebook B (columns) against Codebook A (rows), collapsed to the binary reactant/none dichotomy, matrix sample.

Step 2 – gate consistency (a separate diagnostic). Once Jev says “reactance present”, a downstream model can still answer with `keine_reaktanz`: it *rejects the gate’s premise*. That is not type disagreement, and it must not be mixed into the type analysis. How often each model does this is its own number (Table 5, Figure 12): Jev itself re-labels 48% (16/33) of its own gated comments as `keine_reaktanz` in the type stage; GPT-6-Luna the most (79% (26/33)), DeepSeek-V4.1-Flash 48% (16/33), GLM-5.3-Flash 39% (13/33). Between a fifth and four-fifths of Jev’s gated positives are thus not accepted as reactant by a given second model – a gate-rejection diagnostic that says nothing about which *type* the retainers then choose.

Model	<i>n</i> gated	retain	reject	reject
jev-1.13	33	17	16	48%
gpt-6.luna	33	7	26	79%
deepseek-fl.	33	17	16	48%
glm-5.3-fl.	33	20	13	39%

Table 5. Gate consistency: of the 33 comments Jev’s gate let through (matrix sample, Condition B), how many does each downstream model retain as reactant (assign one of the six types) and how many does it re-label as `keine_reaktanz`?

Step 3 – conditional type classification. The type task, conditioned on the gate, is a **six-class** task: the six reactance types, without `keine_reaktanz`, which at this stage means “you rejected the gate”, not “a kind of reactance”. Figure 13 is the six-class type-confusion grid: only comments on which *both* models of a pair retained the gate enter a cell, and each cell carries its (conditional) *n*, because the support differs from pair to pair once rejections are removed. The type calls on this set are spread out more than one would expect for a task where the answer is almost always the same type: pairwise agreement ranges from 85 to 100% raw with per-pair *n* between 5 and 14 – most pairs still agree on the dominant type (attack), while a few differ on whether the (rare) other types apply at all (κ 0.00–1.00; on the same gated set, the unconditional 7-class task the models were actually prompted with, the same pairs agreed to 58–70%, κ 0.18–0.46). Two further caveats attach to these numbers: the conditional support is tiny (33 comments), and the six types are *sparse* – on the gated set almost every retained type call is `konfrontation_angriff` (attack), with the other five types

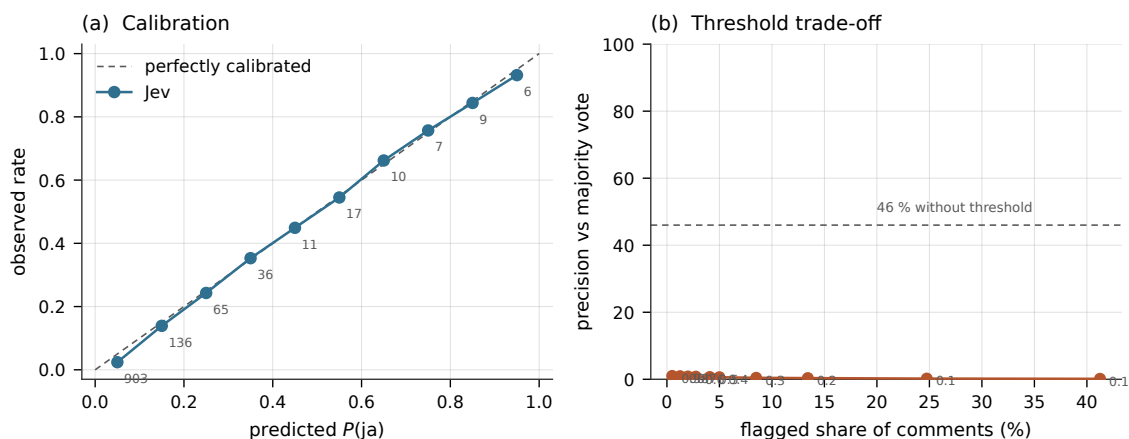


Figure 11. (a) Jev’s $P(ja)$ against the four-model majority vote: observed reactant rate per probability bin versus the predicted probability (grey = the diagonal). This is alignment with a consensus proxy, not calibration against human labels. (b) The precision – coverage trade-off as the threshold t on $P(ja)$ rises; the 46% line is Jev’s threshold-free precision.

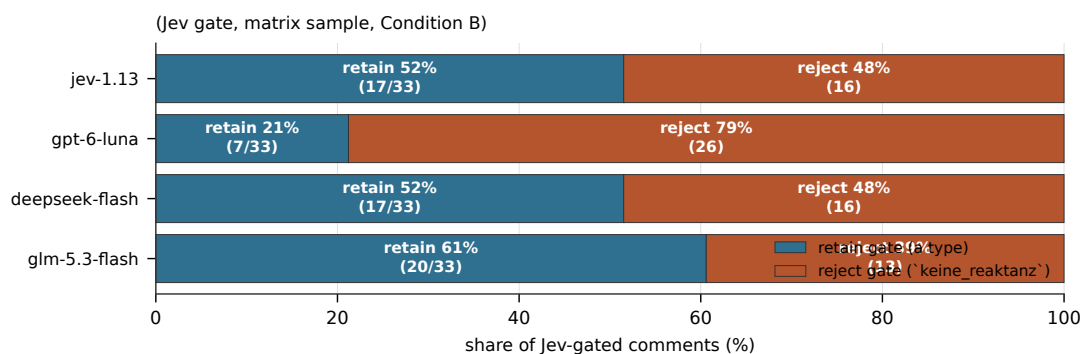


Figure 12. Gate retention vs. rejection per model (matrix sample, Condition B). GPT-6-Luna rejects Jev’s gate most often; GLM-5.3-Flash retains it most often.

at single digits in total (support annotated in Figure 13). A global six-class κ on near-empty classes is therefore not a robust summary; what the data support is a *binary* finding – attack versus the (very rare) other types – plus a statement of the sparsity.

Step 4 – consensus, tie-safe. The majority vote is recomputed on the six types among the models that *retained* the gate, with explicit ties: a 2–2 split is stored as a tie (label null), not resolved by label order, and no model is scored against an arbitrarily tie-broken reference. The two quantities the old single statistic conflated are now reported separately (Figure 14): *gate acceptance* (5 of 33 comments have all four models retaining the gate, 6+5 = 17 have at least three, and 7 are rejected by all) and *conditional type consensus* (among the retainers: 5 four-of-four, 6 three-of-four, 6 two-of-four – where “2 of 4” is a plurality, not a majority – 9 one-of-four, 0 ties). The practical reading does not change, but is now honest: requiring three-or-more models to retain the gate keeps 52% of the gated stream and keeps the type label stable.

On the large-scale sample the gate is even more selective: 58 of 2001 comments (2.90%), of which only 18 reach a plurality of the retaining models on the type. The gate thus concentrates the analysis where it is needed, at roughly 3% of the comments, without losing the ability to compare the models on a common set.

The substantive result of the separation: *the hard problem is not only deciding which reactance type applies. A substantial fraction of the apparent “disagreement” already concerns whether Jev’s gated positives satisfy the reactance definition at all.* Combining both sources in one 7-class matrix overstated

the type problem and understated the gate problem.

6 Two validation experiments

6.1 Repeat stability and position robustness

Identical inputs were coded twice on oversampled positives and a matched negative control. Jev is 98.47% stable overall, with 100.0% on the control and 96.94% on the positives; flips sit at the decision boundary $p(\text{ja}) \approx 0.31$. Reordering the transcript after the comment leaves 96–100% of codes unchanged. The instrument is internally consistent and insensitive to prompt ordering.

6.2 Surface robustness

Three deterministic, meaning-preserving rewrites of the comment: emphasis removed, a neutral politeness frame added, filler particles removed. Emphasis and filler barely move the label; the politeness frame flips 22–38% of positive codes (Table 6). This is not a bug but a hint that the codebook was theoretically not sharp at this boundary — Gate 3 is the consequence.

Stratum	n	emphasis	politeness	filler	all three
3 von 3	27	96%	78%	93%	78%
2 von 3	26	92%	62%	88%	62%
1 von 3	35	97%	77%	89%	69%
Kontrolle	105	100%	98%	100%	98%

Table 6. Surface robustness (Jev, Codebook A, Condition A): share of codes unchanged under each rewrite.

7 Discussion

Prevalence, not effect. Reactance is rare in the studied corpus. That is a finding, not a flaw — and it shifts the work from measuring frequency to validating it. At low prevalence the question is not which model more often says yes, but whether a label carries at all.

The context work is avoidable. Neither transcript nor video measurably affect aggregate detection, and the case-level evidence (Section 2.1) shows the transcript does not carry additional item-level signal either: the flip counts are small in both directions (~ 11 –59 per model on Codebook B, 27 — 121 on Codebook A), and the transcript changes neither the aggregate hit rate nor, case by case, which comments are called reactant. The most practical message for a pipeline over 6.7 million comments is that the comment text suffices and the transcripts remain available for the intervention side of the project.

Cascades instead of monoliths. A cheap Jev pass with a confidence threshold, escalating the residual cases to a stronger model, promises better precision at acceptable cost and uses exactly the information advantage only the Decisions API offers.

A ground truth is still missing. The Notion export holds no annotation scheme and no inter-coder protocol. The re-coding in Section 4.1 and the F1 figures in Section 4.2 are a first step, but were produced by the assistant and replace no supervised double coding with a training phase. The κ - and AC1-values measure consistency among models, not correctness against human coding; “precision” throughout this report is precision against a three-model audit that predates the fourth model, and “F1

against the majority” is an agreement measure. Until a labelled validation set exists, all such estimates are orders of magnitude.

8 Limitations

The sample was built for the method question, not as a representative sample: comments under 25 characters, replies and non-public comments are excluded, and party cells are too small for group comparisons. The transcripts come from automatic speech recognition with substantial errors; Condition A suffers more than Condition B. The precision estimate rests on 36 cases and one coder (the assistant), on the original three-model pool only; the 4-model audit (including the GLM-only flags) is sampled and pending. The majority vote is not a ground truth. The gated six-class analysis runs on 33 conditional observations in the matrix sample, most of them one type: the type-level results there are descriptive, and so is the sparsity. Finally, the confidence bands of the Decisions API are sharply bimodal – 1,521 comments below $p = 0.1$ are never positive, 43 above $p = 0.6$ always are – so the threshold calibrated here is not yet transferable to new data.

A Detailed tables

The full model-pair, condition-A/B and codebook matrices are too large for the main body; they and the per-request logs are regenerable from the repository (Section C).

B Codebook

Codebook A – binary. `ja` iff both (1) a perceived restriction of one’s own freedom (trigger dimensions A–D) and (2) an affective or behavioural counter-reaction. Otherwise `nein`.

Codebook B – seven types. The eight archetypes of the DemocraGPT typology, condensed to disjoint labels:

- `konfrontation_angriff` – destruktiver und konstruktiver Angreifer
- `ablenkung_whataboutism` – aggressiver Ablenker und Ablenkungs-Strategie
- `delegierung_hilflosigkeit` – hilfloser Delegierer
- `vermeidung_rueckzug` – vermeidender Rechtfertiger
- `reflektierte_rechtfertigung` – reflektierter Rechtfertiger
- `konstruktive_kritik` – konstruktiver Kritiker
- `keine_reaktanz` – no reactance

Gates. Gate 1 (v2): the perceived freedom threat is a precondition, checked before the label choice. Gate 2 (v3): the trigger must be the target of the reaction. Gate 3 (v3): politeness markers are neither trigger nor shield. All gates stand in the chat instructions *and* in the Jev criteria.

C Reproduction

```
git clone https://github.com/DanielMatterTUM/democragpt-experiments
cd democragpt-experiments
pip install -r requirements.txt -r requirements-analysis.txt
export OPENROUTER_API_KEY=...
python3 src/build_dataset.py --target 1200 --n-accounts-per-party 8 \
  --out sample_matrix
python3 src/run_benchmark.py --models jev-1.13 gpt-6-luna \
  deepseek-v4.1-flash glm-5.3-flash --codebooks A B \
  --conditions A B --workers 12 --run-name full
python3 src/dedup_one.py requests_full
```

```
# glm on the large-scale sample (condition B only)
python3 src/run_benchmark.py --models glm-5.3-flash --codebooks A B \
    --conditions B --sample sample_big.jsonl --run-name glm_big
# analysis (all four steps before the figures, the report asserts their meta)
python3 src/analyze_extended.py
python3 src/analyze_big.py
python3 src/analyze_gate.py
python3 src/score_audit.py
python3 src/make_audit_sample_v2.py # draws the pending 4-model audit sample
python3 src/exp_reliability.py 45
python3 src/exp_paraphrase.py 35
python3 src/make_report.py
python3 src/figures.py
python3 src/validate_report.py # consistency check, fails the build
python3 src/generate_latex.py
cd report && ./make.sh
```

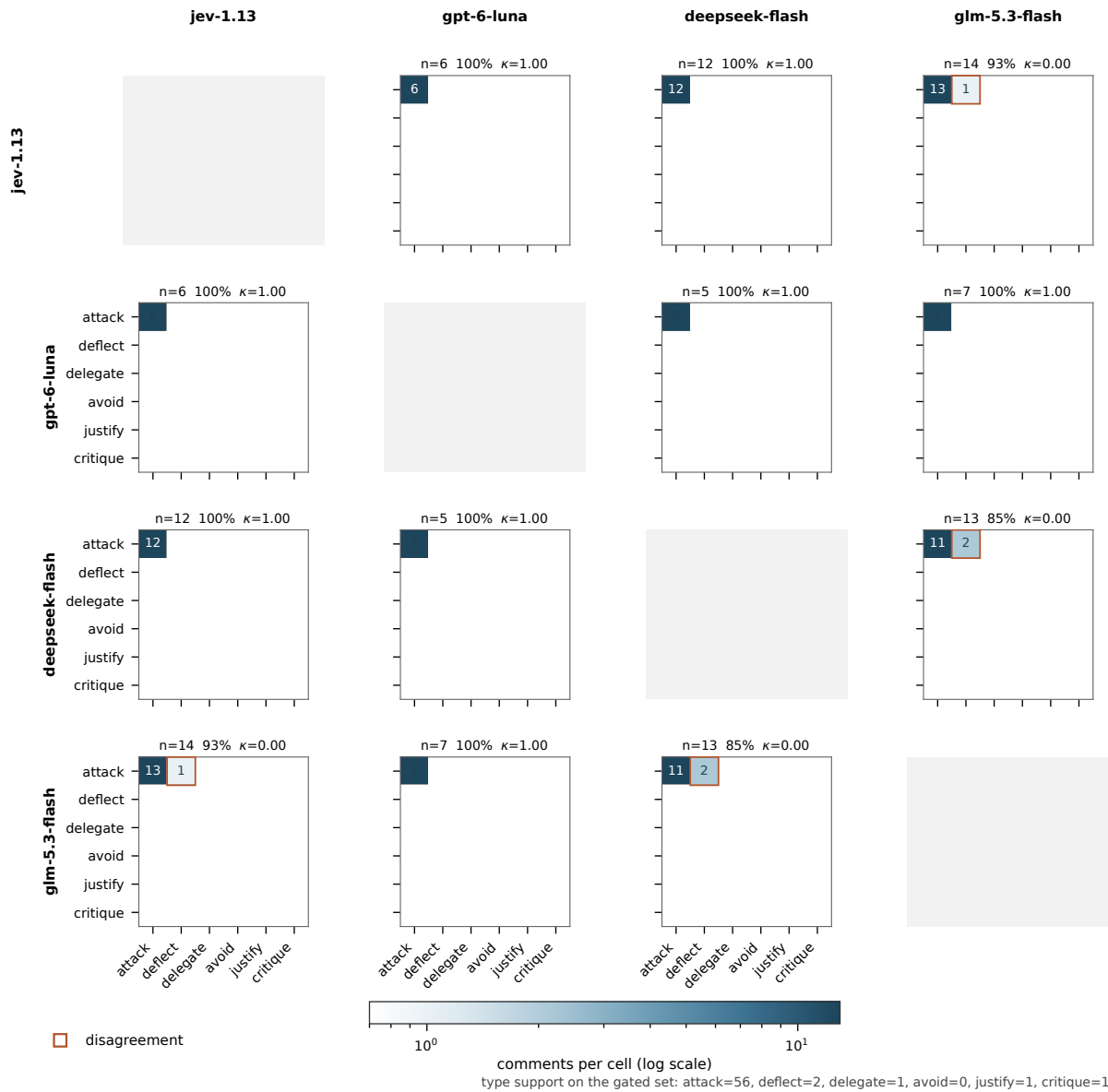


Figure 13. Conditional six-class type confusion for the gated condition (matrix sample, Condition B): rows and columns are the four models' type labels, restricted to the comments on which both models retained Jev's positive gate. "Given that two models both retain the gate, which types do they confuse?" Each cell carries its conditional n ; per-type support is annotated below.

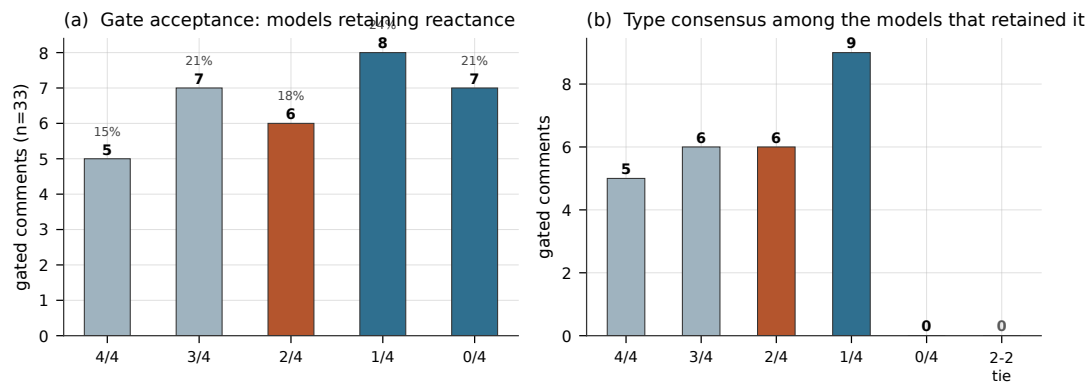


Figure 14. Gated condition, matrix sample, Condition B. (a) Gate acceptance: how many of the four models retain each gated comment's reactance. (b) Conditional type consensus among the models that retained it, with 2-2 ties shown as their own bar.

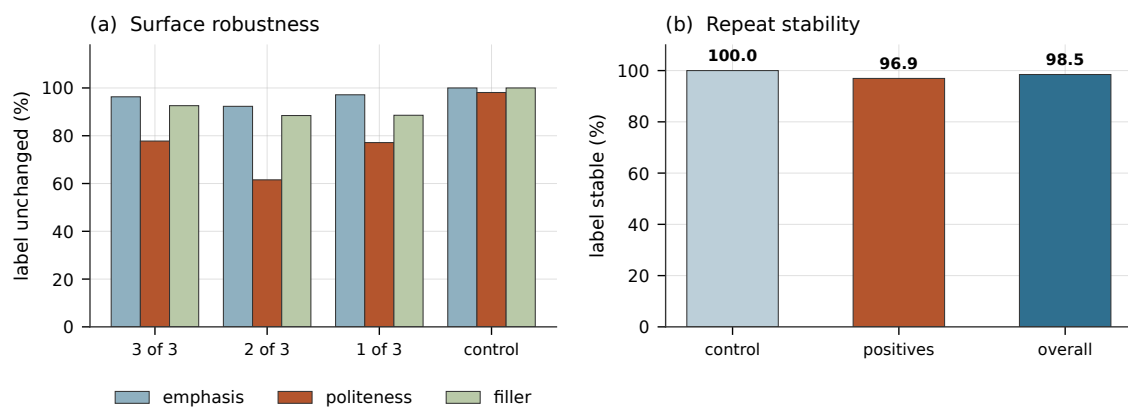


Figure 15. Surface robustness (a) and repeat stability (b), Jev, Codebook A, Condition A.